

Assignment 1 - Data Wrangling: Solution

Introduction to Data Science 11372 & 11516

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Problem description

This report analyses a synthetic daily weather record for a coastal, Sydney-like site from 1 January 2024 to 31 December 2025 (731 days). The record is generated reproducibly from my student ID, so every value reported here is specific to my dataset. It combines instrument-style measurements (temperature, rainfall, humidity, wind, solar exposure, soil moisture and a fire danger class) with the six seasons of the D’harawal calendar.

Objectives

1. Understand the structure and distributions of the data (Part A).
2. Summarise it with vectors, matrices and arrays (Part B).
3. Apply explicit loops and conditional statements (Part C).
4. Reason about, predict, test and debug results against my own data (Part D).

Required libraries

- **knitr**: chunk options and `kable()` tables
- **rmdformats**: the document template

All other analysis uses base R, so no further packages are needed.

Anticipated outcome

A fully reproducible Data Analysis report whose fingerprint (WX-134679-6835-871) matches the marker’s regenerated dataset, with every claim supported by a number in the knitted output.

Setup: Generating the dataset

Run this chunk exactly as provided. **Do not edit the generator.** If you change it, the dataset your marker regenerates will not match yours and your fingerprint check will fail.

```
knitr::opts_chunk$set(echo = TRUE)
## -----
## DO NOT EDIT THIS BLOCK
## -----

seed_from_id <- function(student_id) {
  id <- toupper(gsub("[^0-9A-Za-z]", "", as.character(student_id)))
  if (nchar(id) == 0L) stop("student_id must contain letters or digits.")
  ch <- utf8ToInt(id)
  h <- 5381 # djb2-style rolling hash
  for (k in ch) h <- (h * 33 + k) %% 2147483647
}
```

```

as.integer(h %% 1000000000) + 1L
}

generate_weather_data <- function(student_id,
                                   start_date = as.Date("2024-01-01"),
                                   end_date   = as.Date("2025-12-31")) {

  RNGkind("Mersenne-Twister", "Inversion", "Rejection") # reproducible on R >= 3.6
  set.seed(seed_from_id(student_id))

  dates <- seq.Date(start_date, end_date, by = "day")
  n      <- length(dates)
  doy    <- as.numeric(format(dates, "%j"))
  yr     <- as.integer(format(dates, "%Y"))
  w      <- 2 * pi * (doy - 20) / 365.25                # temperature peak ~20 January

  ## 1. Rainfall: two-state Markov chain (wet and dry spells) -----
  seas_wet <- 1 + 0.18 * cos(2 * pi * (doy - 105) / 365.25) # autumn maximum
  p_dw    <- pmin(0.9, 0.260 * seas_wet)                   # P(wet today | dry yesterday)
  p_ww    <- pmin(0.9, 0.580 * seas_wet)                   # P(wet today | wet yesterday)
  wet     <- logical(n); wet[1] <- runif(1) < 0.3
  for (i in 2:n) wet[i] <- runif(1) < if (wet[i - 1]) p_ww[i] else p_dw[i]
  rain    <- ifelse(wet, rgamma(n, shape = 0.80, scale = 10.5), 0)
  rain    <- round(pmin(rain, 210), 1)
  rain[rain < 0.2] <- 0                                    # below gauge resolution
  wet     <- rain > 0

  ## 2. Temperature: climatology + AR(1) anomaly + summer heatwave spikes ---
  t_clim <- 18.8 + 5.3 * cos(w)
  phi    <- 0.74; sig <- 2.6 * sqrt(1 - phi^2)
  anom   <- numeric(n); anom[1] <- rnorm(1, 0, 2.6)
  for (i in 2:n) anom[i] <- phi * anom[i - 1] + rnorm(1, 0, sig)

  spike <- rbinom(n, 1, 0.012) * rexp(n, 1 / 3.2) * pmax(0, cos(w))^1.5
  heat  <- spike # smear over a 3-day heatwave
  heat[-1] <- heat[-1] + 0.70 * spike[-n]
  heat[-(1:2)] <- heat[-(1:2)] + 0.40 * spike[-((n-1):n)]
  anom <- anom + pmin(heat, 7)

  temp <- t_clim + anom - 1.6 * wet

  ## 3. Relative humidity -----
  rh <- 65 + 2 * cos(w) - 2.1 * anom + 10 * wet + 6 * pmin(1, rain / 20) +
        rnorm(n, 0, ifelse(wet, 4.5, 6.5))
  rh <- pmin(99, pmax(24, rh))

  ## 4. Wind speed -----
  wind <- rweibull(n, shape = 2.1, scale = 15) + 3.5 * wet +
        1.2 * cos(2 * pi * (doy - 300) / 365.25)
  wind <- pmax(1.5, wind)

  ## 5. Solar exposure: clear-sky cycle x cloud factor -----
  ws <- 2 * pi * (doy - 172) / 365.25 # minimum at winter solstice

```

```

clear <- 22 - 10.0 * cos(ws)
cf <- ifelse(wet, rbeta(n, 2.4, 2.6), rbeta(n, 9, 1.4))
solar <- pmax(0.6, clear * cf)

## 6. Soil moisture: single-layer bucket with memory -----
sm <- numeric(n); sm[1] <- 30
for (i in 2:n) {
  et <- (0.12 + 0.035 * max(temp[i], 0)) * (solar[i] / 18) # evapotranspiration
  drain <- 0.032 * (sm[i - 1] - 8) # slow drainage
  sm[i] <- min(max(sm[i - 1] + 0.30 * rain[i] - et - drain, 8), 55)
}
sm <- pmin(56, pmax(5.5, sm + rnorm(n, 0, 0.35)))

## 7. Diurnal range and daily extremes -----
drange <- pmin(14, pmax(2.5, 6.0 + 3.2 * (solar / clear) - 0.03 * rh +
  0.40 * anom + rnorm(n, 0, 0.8)))
tmax <- temp + drange / 2

## 8. Fire danger: McArthur FFDI on afternoon conditions, binned 1-10 -----
drought <- 10 * (1 - (sm - 8) / (55 - 8))
rh_pm <- pmax(8, rh - 18)
wind_pm <- wind * 1.25
ffdi <- 2 * exp(-0.450 + 0.987 * log(pmax(drought, 0.1)) -
  0.0345 * rh_pm + 0.0338 * tmax + 0.0234 * wind_pm)
fri <- as.integer(cut(ffdi,
  breaks = c(-Inf, 1, 2.5, 5, 8, 12, 17, 24, 35, 50, Inf),
  labels = FALSE))

## 9. D'harawal seasons: onsets jittered +/- 10 days each year -----
season_tbl <- data.frame(
  Season = c("Burran", "Marrai'gang", "Burrugin",
    "Wiritjiribin", "Ngonungi", "Parra'dowee"),
  WeatherName = c("Gadalung Marool", "Bana'murrai'yung", "Tugarah Tuli",
    "Tugarah Gunya'marri", "Murray'yunggory", "Goray'murrai"),
  Plant = c("Weetjellat (Lightwood) blooming",
    "Lilly Pilly fruit ripening",
    "Burringoa (Forest Red Gum) flowering",
    "Marrai'uo (Gossamer Wattle) flowering",
    "Miwa Gawaian (Waratah) blooming",
    "Kai'arrewan (Coast Myall) blooming"),
  Animal = c("Male kangaroos aggressive",
    "Marrai'gang (quolls) calling for mates",
    "Echidnas following mating lines",
    "Lyrebirds building display mounds",
    "Flying foxes gathering at dusk",
    "Eels moving downriver to the sea"),
  stringsAsFactors = FALSE)

nominal <- c(91, 152, 213, 244, 305) # nominal onsets, day-of-year
season_idx <- integer(n)
for (y in unique(yr)) {
  b <- sort(nominal + sample(-10:10, 5, replace = TRUE))
  k <- yr == y

```

```

    season_idx[k] <- findInterval(doy[k], b) + 1L
  }

notes <- c(
  "Meat is not eaten during the hottest days; fruit and tubers gathered.",
  "Warm cloaks prepared; groups move toward the coast.",
  "Shellfish are not taken until the boo'kerrikin blooms.",
  "Gentle spring rains begin; lyrebird calls heard at dawn.",
  "Season of ceremony and gathering.",
  "Fish and eels are plentiful in the bays and estuaries.")

df <- data.frame(
  Date           = dates,
  StationID      = "SYD-CICS-01",
  Temperature    = round(temp, 1),
  MaxTemp        = round(tmax, 1),
  MinTemp        = round(temp - drange / 2, 1),
  Rainfall       = rain,
  Humidity        = round(rh, 1),
  WindSpeed      = round(wind, 1),
  SolarExposure  = round(solar, 1),
  SoilMoisture   = round(sm, 1),
  FireRiskIndex  = fri,
  Season         = season_tbl$Season[season_idx],
  SeasonWeatherName = season_tbl$WeatherName[season_idx],
  IndicatorPlant  = season_tbl$Plant[season_idx],
  IndicatorAnimal = season_tbl$Animal[season_idx],
  CommunityNote   = notes[season_idx],
  stringsAsFactors = FALSE)

df$Season <- factor(df$Season, levels = season_tbl$Season)
df
}

dataset_fingerprint <- function(df) {
  num <- unlist(df[vapply(df, is.numeric, logical(1))], use.names = FALSE)
  s <- as.integer(round(sum(num, na.rm = TRUE)))           %% 1000000L
  k <- as.integer(sum(utf8ToInt(paste0(as.character(df$Season),
                                     collapse = ""))))    %% 10000L
  d <- as.integer(which.max(df$MaxTemp) + 7 * which.max(df$Rainfall)) %% 1000L
  sprintf("WX-%06d-%04d-%03d", s, k, d)
}

## -----
## Generate YOUR dataset
## -----

weather_data <- generate_weather_data(params$studentID)

## Untouched copy of the data exactly as generated. Do not modify or delete it.
## The fingerprint must always be computed from this object, never from
## `weather_data`, because later parts of the assignment add columns to
## `weather_data` and that would change the fingerprint.
weather_data_orig <- weather_data

```

```

cat("Student ID :", params$studentID, "\n")

## Student ID : u3284342

cat("Seed      :", seed_from_id(params$studentID), "\n")

## Seed      : 12109058

cat("Rows x Cols :", nrow(weather_data), "x", ncol(weather_data), "\n")

## Rows x Cols : 731 x 16

cat("FINGERPRINT :", dataset_fingerprint(weather_data_orig), "\n")

## FINGERPRINT : WX-134679-6835-871

head(weather_data)

##           Date StationID Temperature MaxTemp MinTemp Rainfall Humidity
## 1 2024-01-01 SYD-CICS-01      23.7    27.4    20.0      0.0    74.7
## 2 2024-01-02 SYD-CICS-01      24.3    27.1    21.5      0.0    71.1
## 3 2024-01-03 SYD-CICS-01      21.0    23.7    18.3      2.3    79.7
## 4 2024-01-04 SYD-CICS-01      20.8    22.3    19.2      2.7    83.9
## 5 2024-01-05 SYD-CICS-01      23.5    26.3    20.8      0.0    79.8
## 6 2024-01-06 SYD-CICS-01      26.4    30.4    22.4      0.0    82.4
## WindSpeed SolarExposure SoilMoisture FireRiskIndex Season SeasonWeatherName
## 1      5.5        31.0        30.2              3 Burran   Gadalong Marool
## 2      8.8        22.3        27.8              3 Burran   Gadalong Marool
## 3      6.7        18.4        27.1              2 Burran   Gadalong Marool
## 4     31.0        10.0        26.9              3 Burran   Gadalong Marool
## 5     26.7        29.5        25.0              4 Burran   Gadalong Marool
## 6      9.5        31.2        22.2              3 Burran   Gadalong Marool
##           IndicatorPlant IndicatorAnimal
## 1 Weetjellan (Lightwood) blooming Male kangaroos aggressive
## 2 Weetjellan (Lightwood) blooming Male kangaroos aggressive
## 3 Weetjellan (Lightwood) blooming Male kangaroos aggressive
## 4 Weetjellan (Lightwood) blooming Male kangaroos aggressive
## 5 Weetjellan (Lightwood) blooming Male kangaroos aggressive
## 6 Weetjellan (Lightwood) blooming Male kangaroos aggressive
##                                     CommunityNote
## 1 Meat is not eaten during the hottest days; fruit and tubers gathered.
## 2 Meat is not eaten during the hottest days; fruit and tubers gathered.
## 3 Meat is not eaten during the hottest days; fruit and tubers gathered.
## 4 Meat is not eaten during the hottest days; fruit and tubers gathered.
## 5 Meat is not eaten during the hottest days; fruit and tubers gathered.
## 6 Meat is not eaten during the hottest days; fruit and tubers gathered.

```

Part A: Data Understanding

Every claim in Part A must be supported by a number that appears in your own knitted output.

Task A.1: Describe the dataset in 120 words

This dataset contains 731 daily records (1 Jan 2024 to 31 Dec 2025) from one Sydney station, SYD-CICS-01. The measured variables are mean, maximum and minimum temperature, rainfall, humidity, wind speed, solar exposure, soil moisture and a 1–10 fire risk index. Mean temperature was 17.9 °C, the hottest maximum reached 37.4 °C, and 2512.4 mm of rain fell, with 433 dry days. Each day is also labelled with one of six D'harawal seasons, plus its weather name, indicator plant, indicator animal and a community note; Burran is the most frequent (172 days). The data supports comparing rainfall and soil moisture across the D'harawal seasons, and examining how soil moisture, humidity and temperature drive fire risk.

Task A.2: Summary statistics

```
#-----#
# used in part A.1 to backup data
cat("Number of days      :", nrow(weather_data), "\n")

## Number of days      : 731
cat("Date range          :", format(range(weather_data$Date), "%d %b %Y"), sep = " ")

## Date range          : 01 Jan 2024 31 Dec 2025
cat("\nMean temperature (°C) :", round(mean(weather_data$Temperature), 1), "\n")

##
## Mean temperature (°C) : 17.9
cat("Highest max temp (°C) :", max(weather_data$MaxTemp), "\n")

## Highest max temp (°C) : 37.4
cat("Total rainfall (mm)   :", round(sum(weather_data$Rainfall), 1), "\n")

## Total rainfall (mm)   : 2512.4
cat("Dry days (0 mm)       :", sum(weather_data$Rainfall == 0), "\n\n")

## Dry days (0 mm)       : 433
cat("Days per season:\n")

## Days per season:
print(table(weather_data$Season))

##
##      Burran  Marrai'gang  Burrugin Wiritjiribin  Ngoonungi  Parra'dowee
##      172      134      129      44      131      121
#-----#

cat("\n--- Summary of the full data frame ---\n")

##
## --- Summary of the full data frame ---
print(summary(weather_data))

##      Date      StationID  Temperature      MaxTemp
##  Min.   :2024-01-01  Length   :731  Min.     : 6.2  Min.     : 7.90
##  1st Qu.:2024-07-01  N.unique : 1    1st Qu.:14.1  1st Qu.:17.10
```

```
## Median :2024-12-31 N.blank : 0 Median :18.2 Median :21.20
## Mean :2024-12-31 Min.nchar: 11 Mean :17.9 Mean :20.92
## 3rd Qu.:2025-07-01 Max.nchar: 11 3rd Qu.:21.4 3rd Qu.:24.75
## Max. :2025-12-31 Max. :32.5 Max. :37.40
## MinTemp Rainfall Humidity WindSpeed
## Min. : 4.60 Min. : 0.000 Min. :33.80 Min. : 1.50
## 1st Qu.:11.20 1st Qu.: 0.000 1st Qu.:63.15 1st Qu.: 9.65
## Median :15.10 Median : 0.000 Median :71.00 Median :14.40
## Mean :14.88 Mean : 3.437 Mean :70.60 Mean :15.07
## 3rd Qu.:18.50 3rd Qu.: 3.950 3rd Qu.:77.90 3rd Qu.:19.70
## Max. :27.70 Max. :67.300 Max. :97.20 Max. :41.70
## SolarExposure SoilMoisture FireRiskIndex Season
## Min. : 1.00 Min. : 7.20 Min. :1.000 Burran :172
## 1st Qu.:10.25 1st Qu.:14.00 1st Qu.:3.000 Marrai'gang :134
## Median :14.80 Median :21.20 Median :3.000 Burrugin :129
## Mean :15.82 Mean :22.21 Mean :3.404 Wiritjiribin: 44
## 3rd Qu.:21.90 3rd Qu.:28.80 3rd Qu.:4.000 Ngoonungi :131
## Max. :31.70 Max. :55.10 Max. :7.000 Parra'dowee :121
## SeasonWeatherName IndicatorPlant IndicatorAnimal CommunityNote
## Length :731 Length :731 Length :731 Length :731
## N.unique : 6 N.unique : 6 N.unique : 6 N.unique : 6
## N.blank : 0 N.blank : 0 N.blank : 0 N.blank : 0
## Min.nchar: 12 Min.nchar: 26 Min.nchar: 25 Min.nchar: 33
## Max.nchar: 19 Max.nchar: 37 Max.nchar: 38 Max.nchar: 69
##
```

```
# (a) mean, sd and sd/mean ratio of every numeric column
num_vars <- weather_data[sapply(weather_data, is.numeric)]
stats <- data.frame(Mean = sapply(num_vars, mean),
                    SD = sapply(num_vars, sd))
stats$SD_to_Mean <- stats$SD / stats$Mean
print(round(stats[order(-stats$SD_to_Mean), ], 3))
```

```
##           Mean      SD SD_to_Mean
## Rainfall    3.437  6.780    1.973
## WindSpeed   15.067  7.238    0.480
## SolarExposure 15.816  7.517    0.475
## SoilMoisture 22.212 10.284    0.463
## FireRiskIndex 3.404  1.154    0.339
## MinTemp     14.877  4.516    0.304
## Temperature 17.902  4.835    0.270
## MaxTemp     20.921  5.262    0.251
## Humidity    70.604  9.921    0.141
```

```
cat("\nLargest SD/mean ratio:", rownames(stats)[which.max(stats$SD_to_Mean)],
    "(", round(max(stats$SD_to_Mean), 3), ")\n")
```

```
##
## Largest SD/mean ratio: Rainfall ( 1.973 )
```

```
# (b) why summary() treats Season differently from Temperature
cat("\nClass of Season      :", class(weather_data$Season), "\n")
```

```
##
## Class of Season      : factor
```

```
cat("Class of Temperature:", class(weather_data$Temperature), "\n")
```

```
## Class of Temperature: numeric
```

```
print(summary(weather_data$Season))
```

```
##      Burran  Marrai'gang  Burrugin Wiritjiribin  Ngoonungi  Parra'dowee
##      172      134      129      44      131      121
```

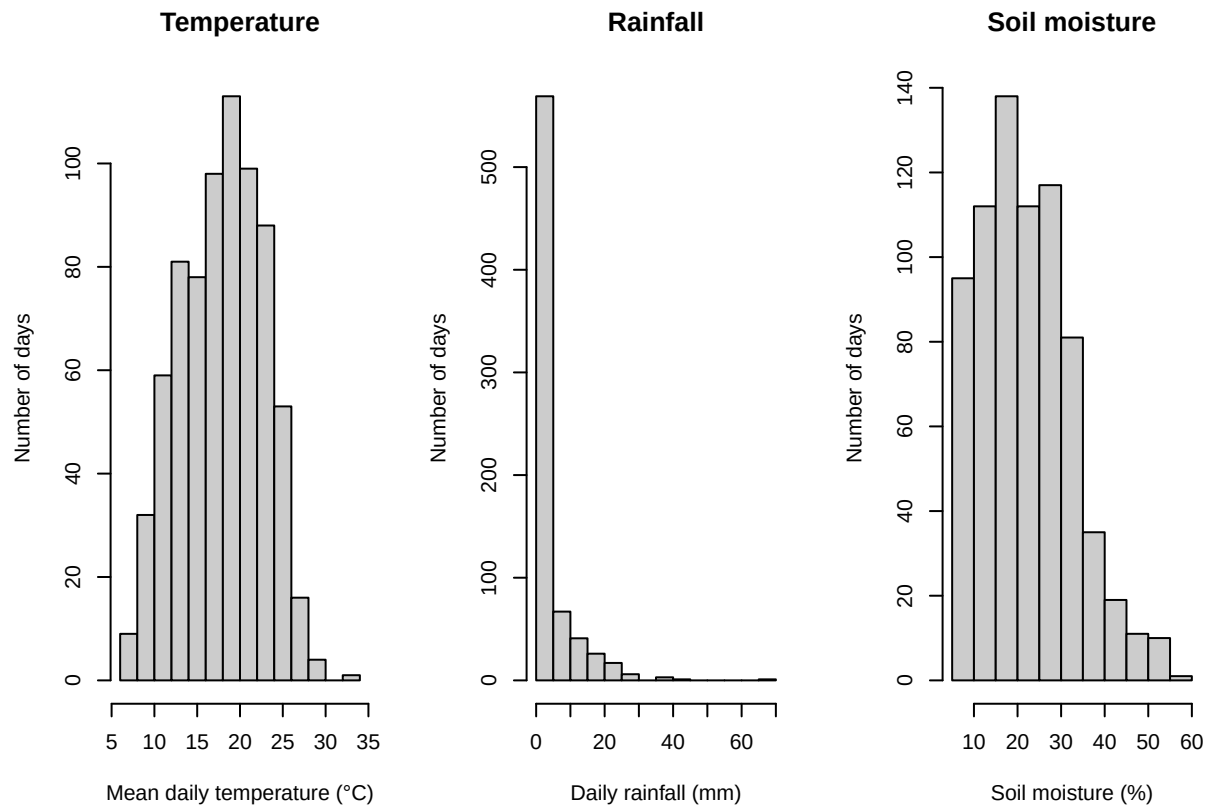
```
print(summary(weather_data$Temperature))
```

```
##      Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
##      6.2   14.1   18.2   17.9   21.4   32.5
```

- The numeric variable with the largest ratio of standard deviation to mean is Rainfall (1.973, an SD of 6.78 mm against a mean of 3.437 mm), which tells me that daily rainfall is extremely variable and right-skewed, with a median of 0 mm and a maximum of 67.3 mm, so a few heavy-rain days supply most of the total and the mean does not describe a typical day.
- `summary()` treats `Season` differently from `Temperature` because `Season` is a factor while `Temperature` is numeric, so it reports the count of days in each level (e.g. Burran 172, Wiritjiribin 44) instead of quadrilles and a mean (17.9 °C for `Temperature`). To get a mean for it I would have to convert it to numeric with `as.numeric(weather_data$Season)`, giving level codes 1–6, although that mean would be meaningless because the seasons are categories, not quantities.

Task A.3: Histograms

```
#-----#
par(mfrow = c(1, 3))
hist(weather_data$Temperature, main = "Temperature",
      xlab = "Mean daily temperature (°C)", ylab = "Number of days", col = "grey80")
hist(weather_data$Rainfall, main = "Rainfall",
      xlab = "Daily rainfall (mm)", ylab = "Number of days", col = "grey80")
hist(weather_data$SoilMoisture, main = "Soil moisture",
      xlab = "Soil moisture (%)", ylab = "Number of days", col = "grey80")
```



```
par(mfrow = c(1, 1))
#-----#
```

Task A.4: Read your own plots (max 150 words total)

```
dry_days <- sum(weather_data$Rainfall == 0)
h_rain <- hist(weather_data$Rainfall, plot = FALSE)
cat("Dry days", dry_days, "\n")
```

```
## Dry days : 433
```

```
cat("First bar", h_rain$breaks[1], "to", h_rain$breaks[2], "mm, height", h_rain$counts[1], "\n")
```

```
## First bar : 0 to 5 mm, height 569
```

```
cat("Difference", h_rain$counts[1] - dry_days, "\n\n")
```

```
## Difference : 136
```

```
season_means <- sapply(weather_data[, c("SoilMoisture", "Rainfall")],
  function(x) tapply(x, weather_data$Season, mean))
round(season_means, 2)
```

```
##           SoilMoisture Rainfall
## Burran      17.71      3.73
## Marrai'gang 24.67      3.78
## Burrugin    32.57      3.65
## Wiritjiribin 33.16      3.88
## Ngoonungi   19.91      2.36
## Parra'dowee 13.37      3.42
```

```
cat("\nHighest SoilMoisture:", names(which.max(season_means[, "SoilMoisture"])), "\n")
```

```
##
```

```
## Highest SoilMoisture: Wiritjiribin
```

```
cat("Highest Rainfall      :", names(which.max(season_means[, "Rainfall"])), "\n")
```

```
## Highest Rainfall      : Wiritjiribin
```

- The shape of my Temperature histogram is broad and slightly left-skewed (mean 17.9 °C, median 18.2 °C), with a steep drop above 25 °C and a thin tail to 32.5 °C, and the data were generated that way because the annual cosine climatology spreads days across a wide range, rain days are cooled by 1.6 °C, and rare summer heatwave spikes form the thin right tail.
- My number of dry days is 433. The height of the first bar of the histogram is 569. The difference of 136 days is made up of wet days with 0.2 to 5 mm of rain.
- The season with the highest mean SoilMoisture is Wiritjiribin (value 33.16 %). It is the same season with the highest mean Rainfall (3.88 mm/day), because its rain adds to water already stored through cool, low-evaporation Burrugin (32.57 %).

Part B: Vectors, Matrices and Arrays

Throughout Part B, “month” means a calendar month within a specific year (2024-01 ... 2025-12), giving 24 groups.

B.1 — avg_temp_month

```
#-----#
```

```
month_id <- format(weather_data$Date, "%Y-%m") # 24 groups: 2024-01 ... 2025-12
```

```
avg_temp_month <- tapply(weather_data$Temperature, month_id, mean)
```

```
print(round(avg_temp_month, 2))
```

```
## 2024-01 2024-02 2024-03 2024-04 2024-05 2024-06 2024-07 2024-08 2024-09 2024-10
##    22.78    21.73    21.12    18.28    15.49    14.44    12.34    11.81    15.60    19.91
## 2024-11 2024-12 2025-01 2025-02 2025-03 2025-04 2025-05 2025-06 2025-07 2025-08
##    20.12    22.67    23.67    23.32    20.48    16.78    15.78    13.08    11.21    10.86
## 2025-09 2025-10 2025-11 2025-12
##    15.62    19.46    20.53    23.04
```

```
cat("\nMaximum:", round(max(avg_temp_month), 2), "°C in", names(which.max(avg_temp_month)), "\n")
```

```
##
```

```
## Maximum: 23.67 °C in 2025-01
```

```
#-----#
```

The maximum of avg_temp_month is 23.67 °C in January 2025 2025-01 .

B.2 — avg_rain_month, avg_hum_month, avg_soil_month

```
#-----#
```

```
avg_rain_month <- tapply(weather_data$Rainfall, month_id, mean)
```

```
avg_hum_month <- tapply(weather_data$Humidity, month_id, mean)
```

```

avg_soil_month <- tapply(weather_data$SoilMoisture, month_id, mean)

round(avg_rain_month, 2)

## 2024-01 2024-02 2024-03 2024-04 2024-05 2024-06 2024-07 2024-08 2024-09 2024-10
##      2.60      6.04      4.74      3.84      5.00      2.35      7.04      3.93      1.82      2.74
## 2024-11 2024-12 2025-01 2025-02 2025-03 2025-04 2025-05 2025-06 2025-07 2025-08
##      0.53      5.79      3.15      2.43      3.55      2.85      3.51      2.62      3.15      2.08
## 2025-09 2025-10 2025-11 2025-12
##      3.70      2.55      1.99      4.32

round(avg_hum_month, 2)

## 2024-01 2024-02 2024-03 2024-04 2024-05 2024-06 2024-07 2024-08 2024-09 2024-10
##      73.15      76.51      71.32      71.84      70.74      66.11      70.39      70.87      66.26      64.67
## 2024-11 2024-12 2025-01 2025-02 2025-03 2025-04 2025-05 2025-06 2025-07 2025-08
##      69.32      73.34      69.45      71.01      74.40      74.17      70.44      69.38      71.82      75.21
## 2025-09 2025-10 2025-11 2025-12
##      67.31      66.82      68.38      71.62

round(avg_soil_month, 2)

## 2024-01 2024-02 2024-03 2024-04 2024-05 2024-06 2024-07 2024-08 2024-09 2024-10
##      17.09      23.72      24.89      22.68      33.25      34.64      45.77      37.62      27.86      14.07
## 2024-11 2024-12 2025-01 2025-02 2025-03 2025-04 2025-05 2025-06 2025-07 2025-08
##      8.78      19.81      12.50      11.65      17.50      16.25      24.65      23.18      29.82      24.00
## 2025-09 2025-10 2025-11 2025-12
##      25.29      12.35      11.12      13.43

#-----#

```

B.3 — Matrix M_month (24 x 4)

```

#-----#
avg_wind_month <- tapply(weather_data$WindSpeed, month_id, mean)

M_month <- cbind(Temperature_C = avg_temp_month,
                 Humidity_pct   = avg_hum_month,
                 WindSpeed_kmh  = avg_wind_month,
                 SoilMoisture_pct = avg_soil_month)
rownames(M_month) <- names(avg_temp_month)      # "2024-01" ... "2025-12"

dim(M_month)

## [1] 24 4

round(M_month, 2)

##      Temperature_C Humidity_pct WindSpeed_kmh SoilMoisture_pct
## 2024-01           22.78          73.15          14.55           17.09
## 2024-02           21.73          76.51          14.47           23.72
## 2024-03           21.12          71.32          15.21           24.89
## 2024-04           18.28          71.84          14.39           22.68
## 2024-05           15.49          70.74          14.45           33.25
## 2024-06           14.44          66.11          14.37           34.64
## 2024-07           12.34          70.39          14.70           45.77

```

```
## 2024-08      11.81      70.87      14.15      37.62
## 2024-09      15.60      66.26      17.63      27.86
## 2024-10      19.91      64.67      14.60      14.07
## 2024-11      20.12      69.32      16.19       8.78
## 2024-12      22.67      73.34      16.20      19.81
## 2025-01      23.67      69.45      15.30      12.50
## 2025-02      23.32      71.01      15.25      11.65
## 2025-03      20.48      74.40      13.80      17.50
## 2025-04      16.78      74.17      13.78      16.25
## 2025-05      15.78      70.44      15.84      24.65
## 2025-06      13.08      69.38      14.47      23.18
## 2025-07      11.21      71.82      15.75      29.82
## 2025-08      10.86      75.21      13.30      24.00
## 2025-09      15.62      67.31      16.49      25.29
## 2025-10      19.46      66.82      15.75      12.35
## 2025-11      20.53      68.38      15.85      11.12
## 2025-12      23.04      71.62      15.16      13.43
```

```
#-----#
```

B.4 — Matrix M_season

```
#-----#
season_vars <- c("Temperature", "Rainfall", "Humidity", "WindSpeed", "SolarExposure")

M_season <- sapply(weather_data[, season_vars],
                  function(x) tapply(x, weather_data$Season, mean))

round(M_season, 1)
```

```
##           Temperature Rainfall Humidity WindSpeed SolarExposure
## Burran             22.3      3.7      72.7      14.9          19.8
## Marrai'gang        16.9      3.8      71.2      14.4          10.9
## Burrugin           12.3      3.7      70.6      14.7           9.2
## Wiritjiribin       11.5      3.9      72.3      13.8          12.0
## Ngoonungi          17.4      2.4      66.3      15.9          18.1
## Parra'dowee        21.6      3.4      70.9      15.9          21.6
```

```
#-----#
```

B.5 — Array A_month (year x month x variable)

```
#-----#
year_id <- format(weather_data$Date, "%Y")
mon_id <- factor(month.abb[as.integer(format(weather_data$Date, "%m"))], levels = month.abb)

array_vars <- c("Temperature", "Rainfall", "Humidity",
               "WindSpeed", "SolarExposure", "FireRiskIndex")

A_month <- array(NA, dim = c(2, 12, length(array_vars)),
                dimnames = list(Year = c("2024", "2025"),
                                Month = month.abb,
                                Variable = array_vars))

for (v in array_vars) {
```

```

  A_month[, , v] <- tapply(weather_data[[v]], list(year_id, mon_id), mean)
}

dimnames(A_month)

## $Year
## [1] "2024" "2025"
##
## $Month
## [1] "Jan" "Feb" "Mar" "Apr" "May" "Jun" "Jul" "Aug" "Sep" "Oct" "Nov" "Dec"
##
## $Variable
## [1] "Temperature" "Rainfall" "Humidity" "WindSpeed"
## [5] "SolarExposure" "FireRiskIndex"
round(A_month[, "Jul", "SolarExposure"], 2)

## 2024 2025
## 8.54 9.16
#-----#

```

The July SolarExposure values are 8.54 in 2024 and 9.16 in 2025.

B.6 — `t(M_month) %*% avg_temp_month`

```

#-----#
B6 <- t(M_month) %*% avg_temp_month
dim(B6)

## [1] 4 1
round(B6, 2)

##           [,1]
## Temperature_C 8114.20
## Humidity_pct 30398.08
## WindSpeed_kmh 6499.86
## SoilMoisture_pct 8868.60
#-----#

```

The units of the first element are $^{\circ}\text{C}^2$, and the result is not a meaningful summary of my data because each element is an uncentred sum of products over the 24 months, not an average or a variance: 8114.20 is simply $24 \times (\text{monthly mean temperature})^2$, so it grows with the number of months, it depends on the arbitrary 0°C zero of the Celsius scale (in Fahrenheit or Kelvin it would change non-proportionally), and the other three elements mix units such as $\% \cdot ^{\circ}\text{C}$ and $\text{km/h} \cdot ^{\circ}\text{C}$ that have no physical meaning.

B.7 — `above_below()` and the TempFlag column

```

#-----#
above_below <- function(temps, months) {
  month_means <- tapply(temps, months, mean)
  sapply(seq_along(temps), function(i) {
    if (temps[i] > month_means[months[i]]) "a" else "b"
  })
}

```

```

weather_data$TempFlag <- above_below(weather_data$Temperature, month_id)

n_a <- sum(weather_data$TempFlag == "a")
cat("Total 'a' days:", n_a, "of", nrow(weather_data), "\n\n")

## Total 'a' days: 365 of 731

prop_a <- tapply(weather_data$TempFlag == "a", month_id, mean)
round(prop_a, 3)

## 2024-01 2024-02 2024-03 2024-04 2024-05 2024-06 2024-07 2024-08 2024-09 2024-10
## 0.645 0.552 0.484 0.500 0.516 0.400 0.452 0.484 0.500 0.484
## 2024-11 2024-12 2025-01 2025-02 2025-03 2025-04 2025-05 2025-06 2025-07 2025-08
## 0.567 0.516 0.516 0.536 0.419 0.500 0.452 0.567 0.484 0.452
## 2025-09 2025-10 2025-11 2025-12
## 0.533 0.484 0.500 0.452

cat("\nHighest proportion:", names(which.max(prop_a)), "(", round(max(prop_a), 3), ")\n")

##
## Highest proportion: 2024-01 ( 0.645 )
#-----#

```

Total number of “a” days: 365 Month with the highest proportion of “a” days: 2024-01 (proportion 0.645)

The total is not exactly half of 731 because 731 is odd, so an exact half (365.5) is impossible, and because each day is compared with its own month’s mean rather than its median, so a month only splits 50/50 when its temperatures are symmetric; in skewed months such as 2024-01 (0.645) and 2024-06 (0.400) the split is uneven, and days exactly equal to the mean count as “b”.

B.8 — Year-on-year comparison

```

#-----#
mm <- sprintf("%02d", 1:12)
yoy <- avg_temp_month[paste0("2025-", mm)] - avg_temp_month[paste0("2024-", mm)]
names(yoy) <- month.name
round(yoy, 2)

## January February March April May June July August
## 0.89 1.58 -0.64 -1.50 0.29 -1.36 -1.13 -0.95
## September October November December
## 0.02 -0.44 0.40 0.37

cat("\nLargest warming:", names(which.max(yoy)), "(", round(max(yoy), 2), "°C)\n")

##
## Largest warming: February ( 1.58 °C)

cat("Largest cooling:", names(which.min(yoy)), "(", round(min(yoy), 2), "°C)\n")

## Largest cooling: April ( -1.5 °C)
#-----#

```

Largest warming: February (1.58 °C). Largest cooling: April (-1.5 °C).

Part C: Looping and Conditional Statements

Each chunk below must use an explicit loop or conditional as described, and must contain **one comment line in your own words saying *why* you structured the loop the way you did** — not what the line does.

C.1 — Mean SoilMoisture for each of the 24 months

```
#-----#
# Why: I pre-allocate the vector and loop over the 24 month labels so each iteration fills one named slot
months <- sort(unique(month_id))           # month_id from B.1
soil_month <- numeric(length(months))
names(soil_month) <- months

for (i in seq_along(months)) {
  soil_month[i] <- mean(weather_data$SoilMoisture[month_id == months[i]])
}

round(soil_month, 2)
```

```
## 2024-01 2024-02 2024-03 2024-04 2024-05 2024-06 2024-07 2024-08 2024-09 2024-10
##    17.09    23.72    24.89    22.68    33.25    34.64    45.77    37.62    27.86    14.07
## 2024-11 2024-12 2025-01 2025-02 2025-03 2025-04 2025-05 2025-06 2025-07 2025-08
##     8.78    19.81    12.50    11.65    17.50    16.25    24.65    23.18    29.82    24.00
## 2025-09 2025-10 2025-11 2025-12
##    25.29    12.35    11.12    13.43
```

```
cat("\nHighest:", names(which.max(soil_month)), "(", round(max(soil_month), 2), "%)\n")
```

```
##
## Highest: 2024-07 ( 45.77 %)
```

```
cat("Lowest :", names(which.min(soil_month)), "(", round(min(soil_month), 2), "%)\n")
```

```
## Lowest : 2024-11 ( 8.78 %)
```

```
#-----#
```

Highest: 45.77 % (month 2024-07). Lowest: 8.78 % (month 2024-11).

C.2 — Days with Temperature above 26 °C

```
#-----#
# Why: I loop over every day and use an if statement as a counter so each day is tested against the 26 °C
hot_count <- 0
for (t in weather_data$Temperature) {
  if (t > 26) hot_count <- hot_count + 1
}

cat("Days above 26 °C:", hot_count, "\n")
```

```
## Days above 26 °C: 21
```

```
cat("Percentage      :", round(100 * hot_count / nrow(weather_data), 2), "%\n")
```

```
## Percentage       : 2.87 %
```

```
#-----#
```

Count of Days with a Temperature above 26 °C 21 days , which is 2.87% of the record.

C.3 — Mean Humidity on days below 15 °C

```
#-----#
```

Why: A single pass with if/else fills both groups at once, so the cold-day and other-day means come f

```
cold_sum <- 0; cold_n <- 0
```

```
other_sum <- 0; other_n <- 0
```

```
for (i in seq_len(nrow(weather_data))) {  
  if (weather_data$Temperature[i] < 15) {  
    cold_sum <- cold_sum + weather_data$Humidity[i]; cold_n <- cold_n + 1  
  } else {  
    other_sum <- other_sum + weather_data$Humidity[i]; other_n <- other_n + 1  
  }  
}
```

```
cat("Days below 15 °C      :", cold_n, "\n")
```

```
## Days below 15 °C      : 214
```

```
cat("Mean humidity (< 15 °C):", round(cold_sum / cold_n, 2), "%\n")
```

```
## Mean humidity (< 15 °C): 72.82 %
```

```
cat("Mean humidity (others) :", round(other_sum / other_n, 2), "%\n")
```

```
## Mean humidity (others) : 69.69 %
```

```
#-----#
```

Mean humidity below 15 °C: 72.82 % (214 days); all other days: 69.69 %. The difference goes in that direction because in the generator humidity rises by 2.1 % for every degree the temperature anomaly falls below normal and by 10 % on rain days, and rain days are also cooled by 1.6 °C, so cold days are disproportionately cool-anomaly and wet days.

C.4 — FireRiskIndex above 4 in Burran and Burrugin

```
#-----#
```

Why: One loop with two separate if conditions counts both seasons in a single pass, so both counts us

```
burran_fire <- 0; burrugin_fire <- 0
```

```
for (i in seq_len(nrow(weather_data))) {  
  if (weather_data$FireRiskIndex[i] > 4) {  
    if (weather_data$Season[i] == "Burran") burran_fire <- burran_fire + 1  
    if (weather_data$Season[i] == "Burrugin") burrugin_fire <- burrugin_fire + 1  
  }  
}
```

```
cat("Burran    days with FireRiskIndex > 4:", burran_fire, "of", sum(weather_data$Season == "Burran"), "
```

```
## Burran    days with FireRiskIndex > 4: 39 of 172
```

```
cat("Burrugin days with FireRiskIndex > 4:", burrugin_fire, "of", sum(weather_data$Season == "Burrugin"), "
```

```
## Burrugin days with FireRiskIndex > 4: 3 of 129
```

```
#-----#
```

Burran: 39 of 172 days (22.7 %).

Burrugin: 3 of 129 days (2.3 %).

Each count is consistent with its D'harawal weather name because the FFDI rises with MaxTemp and with the drought factor derived from low soil moisture: Burran (Gadalung Marool, “hot and dry”) combines summer maximums with dry soil (17.71 %), while Burrugin (Tugarah Tuli, “cold”) has low maximums and wet soil (32.57 %), which keep fire danger low.

C.5 — Total Rainfall matrix (24 months x 6 seasons)

```
#-----#
```

Why: The outer loop walks the months and the inner loop the seasons, so each of the 144 cells is visi

```
season_levels <- levels(weather_data$Season)
rain_mat <- matrix(0, nrow = length(months), ncol = length(season_levels),
                  dimnames = list(Month = months, Season = season_levels))
```

```
for (i in seq_along(months)) {
  for (j in seq_along(season_levels)) {
    rows <- month_id == months[i] & weather_data$Season == season_levels[j]
    rain_mat[i, j] <- sum(weather_data$Rainfall[rows])
  }
}
```

```
print(round(rain_mat, 1))
```

##	Season						
##	Month	Burran	Marra'i'gang	Burrugin	Wiritjiribin	Ngoonungi	Parra'dowee
##	2024-01	80.5	0.0	0.0	0.0	0.0	0.0
##	2024-02	175.2	0.0	0.0	0.0	0.0	0.0
##	2024-03	123.1	23.8	0.0	0.0	0.0	0.0
##	2024-04	0.0	115.3	0.0	0.0	0.0	0.0
##	2024-05	0.0	154.9	0.0	0.0	0.0	0.0
##	2024-06	0.0	5.4	65.0	0.0	0.0	0.0
##	2024-07	0.0	0.0	218.2	0.0	0.0	0.0
##	2024-08	0.0	0.0	0.0	121.8	0.0	0.0
##	2024-09	0.0	0.0	0.0	0.0	54.5	0.0
##	2024-10	0.0	0.0	0.0	0.0	54.1	30.9
##	2024-11	0.0	0.0	0.0	0.0	0.0	15.9
##	2024-12	0.0	0.0	0.0	0.0	0.0	179.6
##	2025-01	97.7	0.0	0.0	0.0	0.0	0.0
##	2025-02	68.0	0.0	0.0	0.0	0.0	0.0
##	2025-03	97.4	12.5	0.0	0.0	0.0	0.0
##	2025-04	0.0	85.6	0.0	0.0	0.0	0.0
##	2025-05	0.0	108.8	0.0	0.0	0.0	0.0
##	2025-06	0.0	0.0	78.5	0.0	0.0	0.0
##	2025-07	0.0	0.0	97.7	0.0	0.0	0.0
##	2025-08	0.0	0.0	11.9	48.7	3.9	0.0
##	2025-09	0.0	0.0	0.0	0.0	110.9	0.0
##	2025-10	0.0	0.0	0.0	0.0	79.0	0.0
##	2025-11	0.0	0.0	0.0	0.0	6.2	53.6
##	2025-12	0.0	0.0	0.0	0.0	0.0	133.8

```

cat("\nCells exactly zero:", sum(rain_mat == 0), "of", length(rain_mat), "\n")

##
## Cells exactly zero: 113 of 144
# Separate zeros where the season never occurs from zeros where it occurred but no rain fell
days_mat <- table(month_id, weather_data$Season)
cat("Zero because season absent :", sum(days_mat == 0), "\n")

## Zero because season absent : 111
cat("Zero despite season present :", sum(rain_mat == 0 & days_mat > 0), "\n\n")

## Zero despite season present : 2
# Transition dates within the record
chg <- which(weather_data$Season[-1] != weather_data$Season[-nrow(weather_data)]) + 1
data.frame(Date = weather_data$Date[chg],
            From = weather_data$Season[chg - 1],
            To = weather_data$Season[chg])

##           Date           From           To
## 1  2024-03-28      Burran  Marrai'gang
## 2  2024-06-06  Marrai'gang    Burrugin
## 3  2024-08-02    Burrugin Wiritjiribin
## 4  2024-09-01 Wiritjiribin    Ngoonungi
## 5  2024-10-30    Ngoonungi  Parra'dowee
## 6  2025-01-01  Parra'dowee      Burran
## 7  2025-03-27      Burran  Marrai'gang
## 8  2025-05-30  Marrai'gang    Burrugin
## 9  2025-08-10    Burrugin Wiritjiribin
## 10 2025-08-24 Wiritjiribin    Ngoonungi
## 11 2025-11-04    Ngoonungi  Parra'dowee
#-----#

```

- a. Cells that are exactly zero: 113 out of 144.
- b. So many cells are zero because each month contains only one or two of the six seasons, so 111 cells are zero simply because that season has no days in that month; only 2 are zero despite the season being present (Burrugin in 2024-08, which covers only 1 August, and Burrugin in 2025-05, which covers only 30–31 May, both dry). The pattern of the non-zero cells tells me that the seasons form a diagonal staircase from Burran (January–March) to Parra'dowee (October–December), with two-season rows only at transition months (2024-03, 2024-06, 2024-10, 2025-03, 2025-11), and 2025-08 even spanning three seasons. Because onsets are jittered each year, the transition months differ between 2024 and 2025 (e.g. Wiritjiribin lasted 1 August to 31 August in 2024 but only 10–23 August in 2025), so the D'harawal seasons do not line up with Gregorian months.
- c. Month containing two seasons: March 2024 (Burran and Marrai'gang); transition date: 28 March 2024.

C.6 — Longest dry spell

```

#-----#
# Why: I track the current run and only replace the best run when a new one is strictly longer, which a
rain <- weather_data$Rainfall
cur_len <- 0; cur_start <- NA

```

```

best_len <- 0; best_start <- NA

for (i in seq_along(rain)) {
  if (rain[i] == 0) {
    if (cur_len == 0) cur_start <- i
    cur_len <- cur_len + 1
    if (cur_len > best_len) {
      best_len <- cur_len
      best_start <- cur_start
    }
  } else {
    cur_len <- 0
  }
}

best_end <- best_start + best_len - 1

cat("Longest dry spell :", best_len, "days\n")

## Longest dry spell : 15 days
cat("Start date      :", format(weather_data$Date[best_start]), "\n")

## Start date       : 2025-05-28
cat("End date        :", format(weather_data$Date[best_end]), "\n")

## End date         : 2025-06-11
cat("SoilMoisture start:", weather_data$SoilMoisture[best_start], "%\n")

## SoilMoisture start: 30.6 %
cat("SoilMoisture end  :", weather_data$SoilMoisture[best_end], "%\n")

## SoilMoisture end   : 18.1 %
#-----#

```

Length: 15 days. Start: 28 May 2025. End: 11 June 2025. SoilMoisture on the first day: 30.6 %; on the last day: 18.1 %. The change goes downward because the bucket model keeps losing water to evapotranspiration and drainage every day while no rain arrives to refill it.

Part D: Reasoning, Prediction and Verification

D.1 — Explain your own code (max 150 words)

I chose Part C task 5. First, `months` is a character vector of the 24 month labels and `season_levels` is a character vector of the 6 season names. `rain_mat` is pre-allocated as a 24×6 numeric matrix of zeros, labelled with those names. The outer loop takes month `i` and the inner loop season `j`; for each pair, `rows` is a logical vector of length 731 that is TRUE where a day falls in both, and the rainfall on those days is summed into cell `[i, j]`. After both loops, `rain_mat` is a 24×6 numeric matrix of rainfall totals in mm. The line `rows <- month_id == months[i] & ...` would break with a different number of rows, because `month_id` was built once in B.1 with 731 elements; if `weather_data` changed length, the two vectors would no longer line up, giving recycled, misaligned or NA totals.

D.2 — Predict, then check

Write your predictions in the chunk below **before running anything**.

PREDICTION (write BEFORE running):

Q1. Roughly how many days in your record have Rainfall above 20 mm?

My prediction: Very few. Most rainy days in a temperate coastal climate bring light to moderate falls, so days above 20 mm should be rare, heavy-rain events. My approximately rough figure: Around 14 days

Q2. Will the median of Rainfall be greater or less than the mean? By roughly how much?

My prediction: My prediction: The median will be less than the mean. I expect most days to have little or no rain, while a few heavy-rain outliers (20 mm or more) pull the mean upward, making the distribution strongly right-skewed. My predicted rough figure: approximately 5% difference

Q3. What proportion of the 731 days in your record are dry (Rainfall == 0)?

My prediction: My prediction: A large proportion, well over half, because I expect the simulated Sydney climate to be fairly dry across most seasons, especially during the summer season, with rain arriving on a minority of days.

Now run the three lines below and reconcile.

```
sum(weather_data$Rainfall > 20) # Q1
```

```
## [1] 28
```

```
c(mean = mean(weather_data$Rainfall),  
  median = median(weather_data$Rainfall)) # Q2
```

```
##      mean      median  
## 3.436936 0.000000
```

```
mean(weather_data$Rainfall == 0) # Q3
```

```
## [1] 0.5923393
```

Q1. Actual 28 days (3.8 %) right that it is low, but the cause is the generator's gamma distribution (shape 0.8), which makes most wet-day amounts small, not the climate.

Q2. Actual mean 3.44 mm, median 0 mm: direction right, but I had not realised the median would be exactly 0, because over half the days are dry.

Q3. Actual 0.592: My prediction was correct but upon some further research into this finding, I found that the main factor for this dryness comes from the Markov chain's long dry spells instead of climate being the only factor.

D.3 — Test four claims against your data

```
#-----#  
# Claim 1: wettest calendar month == month with highest mean soil moisture  
#Claim 1: calendar month (pooled across years) with highest total rain vs highest mean soil moisture  
cal_rain <- tapply(weather_data$Rainfall, mon_id, sum) # mon_id from B.5  
cal_soil <- tapply(weather_data$SoilMoisture, mon_id, mean)  
round(rbind(TotalRain_mm = cal_rain, MeanSoil_pct = cal_soil), 1)
```

```
##           Jan   Feb   Mar   Apr   May   Jun   Jul   Aug   Sep   Oct   Nov  
## TotalRain_mm 178.2 243.2 256.8 200.9 263.7 148.9 315.9 186.3 165.4 164.0 75.7  
## MeanSoil_pct  14.8  17.8  21.2  19.5  28.9  28.9  37.8  30.8  26.6  13.2 10.0  
##           Dec
```

```
## TotalRain_mm 313.4
## MeanSoil_pct 16.6
cat("Wettest month:", names(which.max(cal_rain)),
    "| Highest soil moisture:", names(which.max(cal_soil)),
    "| Claim 1:", names(which.max(cal_rain)) == names(which.max(cal_soil)), "\n\n")
```

```
## Wettest month: Jul | Highest soil moisture: Jul | Claim 1: TRUE
```

```
#-----#
# Claim 2: 2025 warmer on average than 2024
# Claim 2: 2025 warmer on average than 2024
yr_temp <- tapply(weather_data$Temperature, year_id, mean)
round(yr_temp, 3)
```

```
## 2024 2025
## 18.014 17.788
```

```
cat("Claim 2:", yr_temp["2025"] > yr_temp["2024"], "\n\n")
```

```
## Claim 2: FALSE
```

```
#-----#
# Claim 3: fewer than 50 days with Temperature above 25 °C
# Claim 3: fewer than 50 days with mean Temperature above 25 °C
n25 <- sum(weather_data$Temperature > 25)
cat("Days above 25 °C:", n25, "| Claim 3:", n25 < 50, "\n\n")
```

```
## Days above 25 °C: 40 | Claim 3: TRUE
```

```
#-----#
# Claim 4: longest run of consecutive dry days longer than 18 days
# Claim 4: longest dry spell longer than 18 days (best_len from C.6)
cat("Longest dry spell:", best_len, "days | Claim 4:", best_len > 18, "\n")
```

```
## Longest dry spell: 15 days | Claim 4: FALSE
```

```
#-----#
```

Claim 1: wettest calendar month is July (315.9 mm) and the highest mean soil moisture is also July (37.8 %) -> TRUE. Claim 2: 2024 mean 18.01 °C, 2025 mean 17.79 °C -> FALSE (2025 was 0.23 °C cooler). Claim 3: 40 days had a mean Temperature above 25 °C -> TRUE. Claim 4: the longest dry spell is 15 days -> FALSE.

Claim 1 can be false even though rainfall is the only source of soil moisture because soil moisture is a bucket with memory, so it reflects rain accumulated over previous weeks and can peak after the wettest month rather than during it. It is also drained by evapotranspiration that rises with temperature and solar exposure, so a wet but hot month such as December (313.4 mm, yet only 16.6 % soil moisture) loses water far faster than a cool winter month.

Claims 2 and 3 came out differently for different students because every dataset uses a different random seed, and the generator's climatology is identical in both years, so the year-to-year difference and the count of hot days are driven purely by each student's random anomalies and heatwave spikes.

D.4 — `classify_day()` with thresholds from your own distribution

```
#-----#
cuts <- quantile(weather_data$Temperature, c(0.20, 0.80))
cuts
```

```
## 20% 80%
```

```
## 13.2 22.6
```

```
classify_day <- function(temps, lo, hi) {  
  sapply(seq_along(temps), function(i) {  
    if (temps[i] > hi) "Hot" else if (temps[i] < lo) "Cool" else "Mild"  
  })  
}  
  
weather_data$DayClass <- classify_day(weather_data$Temperature, cuts[1], cuts[2])  
table(factor(weather_data$DayClass, levels = c("Cool", "Mild", "Hot")))
```

```
##  
## Cool Mild Hot  
## 146 443 142
```

```
#-----#
```

Quantile cut-points used: 13.2 °C (20th percentile) and 22.6 °C (80th percentile), from `quantile()`.

Counts: Hot 142, Mild 443, Cool 146.

A quantile-based threshold is better than a fixed one on another site's data because it adapts to that site's own distribution, always labelling roughly its coolest and hottest 20 % of days, whereas fixed 20 °C and 30 °C thresholds could class almost every day at a cold or tropical site the same way.

D.5 — Debugging: code that runs but is wrong

Both lines use `weather_data_orig`, the unmodified data frame from the setup chunk.

```
# (i)  
avg_temp <- sum(weather_data_orig$Temperature) / length(weather_data_orig)  
avg_temp
```

```
## [1] 817.8813
```

```
# (ii) intended: the mean humidity on hot, dry days  
hot_dry <- mean(weather_data_orig$Humidity[weather_data_orig$Temperature > 25 |  
  weather_data_orig$Rainfall == 0])  
hot_dry
```

```
## [1] 65.65905
```

Now compute the correct versions, and the number of days each version of line (ii) selects.

```
#-----#  
# (i)  
avg_temp_wrong <- sum(weather_data_orig$Temperature) / length(weather_data_orig)  
avg_temp_right <- sum(weather_data_orig$Temperature) / nrow(weather_data_orig)  
cat("(i) wrong:", round(avg_temp_wrong, 2), "| correct:", round(avg_temp_right, 2),  
  "| divided by", length(weather_data_orig), "instead of", nrow(weather_data_orig), "\n\n")
```

```
## (i) wrong: 817.88 | correct: 17.9 | divided by 16 instead of 731
```

```
# (ii)  
sel_or <- weather_data_orig$Temperature > 25 | weather_data_orig$Rainfall == 0  
sel_and <- weather_data_orig$Temperature > 25 & weather_data_orig$Rainfall == 0  
cat("(ii) wrong (|):", round(mean(weather_data_orig$Humidity[sel_or]), 2), "% from", sum(sel_or), "days",  
  "\n\n")
```

```
## (ii) wrong (|): 65.66 % from 442 days
```

```
cat("(ii) right (&):", round(mean(weather_data_orig$Humidity[sel_and]), 2), "% from", sum(sel_and), "days")
```

```
## (ii) right (&): 61.26 % from 31 days
```

```
#-----#
```

Line (i): wrong value 817.88; correct value 17.90.

How did the error occur: `length()` on a data frame returns its number of columns (16), not rows (731), so it divided the total temperature by 16, and the mistake is silent because a data frame is a list of columns, so `length()` is a perfectly valid call that returns a number.

Line (ii): wrong value 65.66 % (selects 442 days); correct value 61.26 % (selects 31 days, using `&`).

How did the error occur: `|` selects days that are hot OR dry, which pulls in all 433 dry days whether hot or not, instead of days that are hot AND dry, and the mistake is silent because `|` and `&` are both valid logical operators that return a logical vector of the same length.

D.6 — Interpretation grounded in your own results

```
#-----#
# (a) rank by mean Temperature, with days per season
season_days <- table(weather_data$Season)
rank_tbl <- data.frame(Season = rownames(M_season),
                      MeanTemp = round(M_season[, "Temperature"], 2),
                      Days = as.integer(season_days[rownames(M_season)]))
rank_tbl[order(-rank_tbl$MeanTemp), ]
```

```
##           Season MeanTemp Days
## Burran          Burran   22.25  172
## Parra'dowee     Parra'dowee  21.59  121
## Ngoonungi       Ngoonungi   17.45  131
## Marrai'gang     Marrai'gang   16.94  134
## Burrugin        Burrugin    12.31  129
## Wiritjiribin    Wiritjiribin  11.45   44
```

```
# (b) "dry" half of Burran as a rate
rain_rate <- tapply(weather_data$Rainfall, weather_data$Season, mean)
rain_prop <- tapply(weather_data$Rainfall > 0, weather_data$Season, mean)
rain_tot <- tapply(weather_data$Rainfall, weather_data$Season, sum)
round(data.frame(MeanRain_mm_per_day = rain_rate,
                 PropRainDays = rain_prop,
                 TotalRain_mm = rain_tot), 3)
```

```
##           MeanRain_mm_per_day PropRainDays TotalRain_mm
## Burran                   3.732         0.442        641.9
## Marrai'gang              3.778         0.485        506.3
## Burrugin                 3.653         0.395        471.3
## Wiritjiribin             3.875         0.386        170.5
## Ngoonungi                2.356         0.298        308.6
## Parra'dowee              3.420         0.413        413.8
```

```
#-----#
```

a. Seasons ranked by mean Temperature, with the number of days each covers:

1. Burran (22.25 °C, 172 days)
2. Parra'dowee (21.59 °C, 121 days)

3. Ngoonungi (17.45 °C, 131 days)
 4. Marrai'gang (16.94 °C, 134 days)
 5. Burrugin (12.31 °C, 129 days)
 6. Wiritjiribin (11.45 °C, 44 days)
- b. “Hot”: supported, as Burran is the hottest season at 22.25 °C. “Dry” tested as a rate: Burran averages 3.73 mm per day with rain on 44.2 % of days, the third-highest daily rate and second-highest rain-day proportion of the six seasons, far wetter than Ngoonungi (2.36 mm per day, 29.8 % rain days). My dataset does not support both halves: it is hot but not dry. A seasonal total would mislead here because Burran covers the most days (172, from part a), so its total of 641.9 mm is inflated by the season’s length, not by how wet each day is.
- c. My dataset cannot be used as evidence for or against the Parra'dowee note because the weather values were simulated by the generator rather than observed on Country, and the CommunityNote is just a fixed label attached to every Parra'dowee day, so nothing in the data measures fish or eels. Testing the statement would require ecological observations such as multi-year fish and eel surveys or catch records from local bays and estuaries, interpreted with guidance from D'harawal Knowledge holders.

D.7 — Write your own classifier

```
#-----#
wet_median <- median(weather_data$Rainfall[weather_data$Rainfall > 0])
wind_q90   <- quantile(weather_data$WindSpeed, 0.90)
cat("Median wet-day rainfall:", wet_median, "mm | 90th percentile wind:", round(wind_q90, 1), "km/h\n\n")

## Median wet-day rainfall: 5.8 mm | 90th percentile wind: 24.7 km/h

classify_weather <- function(rain, wind, rain_cut, wind_cut) {
  ifelse(rain == 0, "Dry",
    ifelse(rain >= rain_cut & wind >= wind_cut, "Stormy", "Wet"))
}

weather_data$WeatherClass <- classify_weather(weather_data$Rainfall, weather_data$WindSpeed,
  wet_median, wind_q90)

table(weather_data$WeatherClass)

##
##      Dry Stormy      Wet
##    433      21    277

# Stormiest day: highest combined standardised rain + wind among Stormy days
storm_score <- as.numeric(scale(weather_data$Rainfall) + scale(weather_data$WindSpeed))
storm_score[weather_data$WeatherClass != "Stormy"] <- NA
s <- which.max(storm_score)
weather_data[s, c("Date", "Rainfall", "WindSpeed", "MaxTemp")]

##           Date Rainfall WindSpeed MaxTemp
## 196 2024-07-14     35.5      29.3     11.2

# Candidate disagreements
not_stormy <- weather_data$WeatherClass != "Stormy"
weather_data[not_stormy, c("Date", "Rainfall", "WindSpeed", "WeatherClass")][
  which.max(weather_data$Rainfall[not_stormy]), ] # heaviest rain not Stormy

##           Date Rainfall WindSpeed WeatherClass
##  71 2024-03-11     67.3      6.5             Wet
```

```
dry <- weather_data$WeatherClass == "Dry"
weather_data[dry, c("Date", "Rainfall", "WindSpeed", "WeatherClass")][
  which.max(weather_data$WindSpeed[dry]), ] # windiest Dry day

##           Date Rainfall WindSpeed WeatherClass
## 701 2025-12-01         0       39.2         Dry
#-----#
```

Thresholds and the statistic justifying each: - Dry: exactly 0 mm, because 433 of my 731 days (59 %) sit at the gauge-resolution floor. - Wet: any rain above 0 mm that does not meet the Stormy rule. - Stormy: rainfall at or above my median wet-day amount (5.8 mm) AND wind at or above my 90th percentile (24.7 km/h), so it captures days that are both genuinely wet and in the windiest 10 %. Counts: Wet 277, Dry 433, Stormy 21. Stormiest date: 14 July 2024: Rainfall 35.5 mm, WindSpeed 29.3 km/h, MaxTemp 11.2 °C. One day I disagree with: 11 March 2024, because it had my record rainfall of 67.3 mm but only 6.5 km/h of wind, so it is labelled Wet even though rain that heavy would be a storm on its own. I would change the rule so that rainfall above my 99th percentile counts as Stormy regardless of wind.

Here's just some code for the results summary table for values that I haven't all ready outputted previously

```
weather_data_orig$Date[which.max(weather_data_orig$MaxTemp)]; max(weather_data_orig$MaxTemp)

## [1] "2025-01-08"
## [1] 37.4

round(cor(weather_data_orig$SoilMoisture, weather_data_orig$FireRiskIndex), 2)

## [1] -0.67
```

Verification and Reproducibility

Results Summary Table

Fill in this table **manually** from your own output above. Rows 1-11 are checked automatically against your student ID; row 12 is marked by inspection. Leave no cell blank; write NA and explain if you could not compute one.

#	Quantity	Your value
1	Dataset fingerprint (from the setup chunk)	WX-134679-6835-871
2	Mean Temperature over the whole record (2 dp)	17.90
3	Total Rainfall over the whole record (1 dp)	2512.4
4	Number of days with Rainfall == 0	433
5	Date of the highest MaxTemp , and that value	2025-01-08, 37.4 °C
6	Season with the highest mean Temperature , and that mean (2 dp)	Burran, 22.25 °C

#	Quantity	Your value
7	Number of days with FireRiskIndex > 4 in Burran	39
8	Length of the longest dry spell (Part C task 6)	15
9	Start date of the longest dry spell (earliest run, if tied)	2025-05-28
10	cor(SoilMoisture, FireRiskIndex) (2 dp)	-0.67
11	Number of "a" days from Part B task 7	365
12	Your two classify_day() quantile cut-points (D.4)	13.2 °C (20th percentile) and 22.6 °C (80th percentile)

Reproducibility appendix

Leave this chunk unchanged as the **last chunk** of your submission.

```
cat("Student ID :", params$studentID, "\n")

## Student ID : u3284342

cat("Fingerprint :", dataset_fingerprint(weather_data_orig), "\n")

## Fingerprint : WX-134679-6835-871

cat("Knitted at :", format(Sys.time(), "%Y-%m-%d %H:%M:%S %Z"), "\n")

## Knitted at : 2026-09-29 03:09:45 AEST

cat("Chunks run :", length(knitr::all_labels()), "\n\n")

## Chunks run : 28

print(sessionInfo()$R.version$version.string)

## [1] "R version 4.6.1 (2026-06-24)"
```

Declaration

I confirm that the dataset analysed in this submission was generated from my own student ID, that the values reported in the Results Summary Table are the values produced by the code in this document, and that all explanations and interpretations are my own work.

Signed: Ruhan Shafi Date: 29 September 2026

Overall Conclusion

Your overall reflection about this assignment goes here ...

Findings

My dataset behaves like a coastal temperate climate: a mean temperature of 17.90 °C with a clear annual cycle (monthly means from 10.86 °C in August 2025 to 23.67 °C in January 2025), and 2512.4 mm of rain concentrated on only 41 % of days, which makes Rainfall by far the most variable variable (SD/mean 1.97).

Soil moisture behaves as a lagged store rather than a mirror of rainfall: December received almost as much rain as July (313.4 mm vs 315.9 mm) yet held less than half the soil moisture (16.6 % vs 37.8 %) because of summer evapotranspiration. Fire danger followed hot days and dry soil, so Burran recorded 39 high-risk days against only 3 in Burrugin. Burran fitted the “hot” half of Gadalung Marool but not the “dry” half, a reminder that a synthetic record calibrated to a climate cannot test Knowledge that is grounded in ecological indicators.

Challenges

Challenge 1: The TempFlag error

My first challenge came from the difference between working in the console and knitting the document. While writing my Part A answers, I ran `summary(weather_data)` in the console after I had already completed Part B, and the output included an extra `TempFlag` column that the Part A questions knew nothing about. At first this was confusing, because the dataset is meant to have 16 columns. The cause was that my console session had kept the column created by `above_below()` in B.7, so the object I was summarising was no longer the dataset as generated.

This showed me that R objects change as the analysis progresses, and that running chunks out of order gives results the knitted document will not reproduce. It also explained why the assignment insists that the fingerprint is computed from `weather_data_orig`: once I add columns such as `TempFlag` to `weather_data`, any calculation over every numeric column would change, and the marker’s regenerated dataset would no longer match mine. To stop this happening again, I treated the knitted output as the only authoritative record, re-knitting from a clean session rather than trusting values left over in the console.

Challenge 2: Recognising when my conclusions were wrong

My second and more significant challenge was recognising when a conclusion I had reached was wrong, whether from incorrect intuition, an error in logic, or jumping to an answer before checking the evidence. In each case the conclusion felt reasonable, so the mistake was not obvious until I compared it with a number in my own output.

The clearest case of incorrect intuition was my D.2 prediction. I predicted a large proportion of dry days because I believed Sydney has a primarily dry climate. The proportion of dry days was indeed high (0.592), but my reasoning was wrong: my own data recorded about 1,256 mm of rain per year, which is consistent with a fairly wet coastal city. The dry days come from the generator’s two-state Markov chain, which produces rain in spells separated by long dry runs, not from low annual rainfall. Getting the right answer for the wrong reason taught me that a correct prediction does not prove my understanding was correct.

An error in logic appeared in B.7, where I first explained that the number of “a” days (365) was not exactly half of 731 because 2024 was a leap year. On reflection, this does not hold: the leap year only determines how many days are in the record, and has no bearing on whether a day is above or below its month’s mean. The real reason is that each day is compared with its monthly mean rather than its median, so a month only splits evenly when its temperatures are symmetric. My monthly proportions, ranging from 0.400 in 2024-06 to 0.645 in 2024-01, show this directly. My first explanation described a true fact about the data that simply did not cause the result.

I also jumped to conclusions before doing the research. In A.4, knowing that soil moisture lags rainfall in the bucket model, I expected the season with the highest mean SoilMoisture to differ from the season with the highest mean Rainfall, and began explaining the mechanism that would separate them. My output showed that Wiritjiribin led both (33.16 % and 3.88 mm per day), so I had to change my answer to explain why they coincided.

Together, these experiences changed how I approached the rest of the assignment. Instead of writing an explanation and then looking for numbers to support it, I began checking the output first and only then explaining what it showed. When a result surprised me, I treated that as a sign that my understanding needed revising rather than as a mistake in the data.

Reflection

The biggest lesson that I learnt from this assessment came from learning to recognising my own wrong conclusions. My D.2 prediction showed that reaching the right answer does not mean my reasoning was right, and my first B.7 explanation showed that a true fact (the leap year) can still be the wrong cause. Jumping ahead in A.4 and D.6 showed that an explanation written before checking the evidence tends to fit what I expected rather than what the data show. Going forward, I will write down my expectations before running code, let the printed output lead the explanation, and treat a surprising result as a sign that my understanding needs revising rather than as an error in the data.

Finally, I am more aware of the limits of a dataset. However realistic a synthetic record is, it cannot be used to test Knowledge that is grounded in observation from real world data, and recognising what data can and cannot support is as important as analysing it correctly.